AI Agent Integration Plan for Seismic Command Dashboard

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1 Project Overview

1.1 Website Topic and Purpose

Seismic Command Dashboard is an AI-powered earthquake monitoring and awareness platform designed for everyday citizens in Turkey. The system fetches real-time earthquake data from Kandilli Observatory (KOERI), visualizes active fault lines from MTA (General Directorate of Mineral Research and Exploration) and soil site classification zones on an interactive map, and provides AI-driven natural language analysis through Groq LLM integration.

The primary goal is to transform complex seismic data into clear, understandable information that helps ordinary people understand earthquake risks, learn how to prepare, and make informed decisions about their safety.

1.2 Target Users

User Group	Use Case
Curious Citizens	Checking recent earthquakes, asking "Was that an earthquake?", understanding what happened
Homeowners & Renters	Analyzing their home's risk: fault proximity, soil class, building safety
Students & Educators	Exploring seismic data, learning about faults and earthquake science
Parents & Families	Earthquake preparedness, emergency bag checklist, teaching children about earthquakes
Travelers	Checking regional seismic risk before visiting a new area

Table 1: Target user groups and their primary use cases.

1.3 Core Features of the Website

1. **Real-Time Earthquake Feed:** Live earthquake data from Kandilli Observatory with 30-second cache, filterable by magnitude (M1+–M5+), depth, and time window (24h–168h).
2. **Interactive Map:** Leaflet/MapLibre-based map with color-coded earthquake markers, MTA active fault line overlays, and soil site classification zones (ZA–ZF).
3. **Dashboard Analytics:** KPI cards (event count, max/avg magnitude, critical events), 7-day trend charts, magnitude distribution histogram, top affected cities, and hourly anomaly detection.
4. **AI Chat Assistant:** Natural language Q&A in Turkish powered by Groq LLM (LLaMA 3.3 70B) for earthquake explanations, fault analysis, preparedness guidance, and risk assessment.
5. **Geospatial Filtering:** Location search, multi-parameter filtering, and dynamic viewport-based data loading.

2 AI Agent Concept

2.1 What Problem Will the AI Agent Solve?

Turkey is located in one of the most seismically active regions in the world. Dozens of earthquakes occur every day, yet ordinary citizens face the following challenges:

1. **Data Interpretation Gap:** Raw earthquake data (magnitude, depth, coordinates) is meaningless to everyday users. They need plain-language explanations like "Was this earthquake dangerous?" or "What does 4.2 magnitude mean for me?"
2. **Fault Line Unawareness:** Most citizens have no idea which fault line runs near their home, what its history is, or what risk it poses. The AI agent bridges this knowledge gap by explaining fault characteristics in simple language.
3. **Preparedness Gap:** People lack practical knowledge about earthquake preparedness — what to put in an emergency bag, what to do during shaking, and how to act after an earthquake.
4. **Home Risk Uncertainty:** Homeowners want to know "Is my home safe?" but cannot interpret soil classification zones (ZA–ZF), Vs30 values, or fault proximity data. The AI translates technical data into actionable advice.
5. **Post-Earthquake Anxiety:** After feeling an earthquake, people panic and search for reliable information. The AI agent provides instant, data-backed answers about what just happened.
6. **Building Safety Concerns:** People living in older buildings or high-rise apartments want to understand their structural risk but lack engineering knowledge to assess it.

2.2 What Type of Agent Will It Be?

The AI Agent is designed as a **multi-role assistant** for everyday users. It combines five functional roles:

Role	Agent Type	Example Interaction
Earthquake Explainer	Assistant	"What just happened? A 4.5 earthquake — was it dangerous?"
Fault Analyst	Advisor	"Which fault line caused this? What's its history?"
Home Risk Assessor	Evaluator	"Is my home safe? Nearest fault is 3km, soil class ZD"
Preparedness Guide	Guide	"What should I put in my earthquake bag?"
Statistics Reporter	Recommender	"How many earthquakes occurred in Turkey last month?"

Table 2: AI Agent roles and example user interactions.

2.3 How Users Will Interact with the Agent

Users will interact with the AI Agent through three primary channels:

2.3.1 1. Chat Interface (Map Page – AI Chat Sidebar)

The main interaction channel is the AI Chat Sidebar on the Map page. Users type free-form questions in Turkish, and the system automatically enriches each request with context (selected earthquake, current map viewport, active filters).

Supported question categories:

- **Earthquake explanation:** "Bu deprem tehlikeli miydi?" (Was this earthquake dangerous?)

- **Fault analysis:** "Bu deprem hangi fay hatti uzerinde oldu?" (Which fault line was this earthquake on?)
- **Fault history:** "Kuzey Anadolu Fay Hatti'nda gecmiste ne depremler oldu?" (What earthquakes occurred on the North Anatolian Fault?)
- **Home risk:** "Evim fay hattina yakin mi? Zemini nasil?" (Is my home close to a fault? How is the soil?)
- **Preparedness:** "Deprem cantasinda neler olmali?" (What should be in an earthquake bag?)
- **Safety during earthquake:** "Deprem aninda ne yapmaliyim?" (What should I do during an earthquake?)
- **Building safety:** "30 yillik 5 katli binada oturuyorum, risk nedir?" (I live in a 30-year-old 5-story building, what is the risk?)
- **Statistics:** "Son 1 ayda Turkiye'de kac deprem oldu?" (How many earthquakes in Turkey last month?)
- **For children:** "Cocuklara depremi nasil anlatmaliyim?" (How should I explain earthquakes to children?)

Three **quick preset buttons** are also provided:

1. **"Hangi fay?"** – Nearest fault with distance and confidence level
2. **"Fay detay"** – Fault slip type, movement history, and velocity info
3. **"Fay aktivitesi"** – Active faults in the current viewport with event counts

2.3.2 2. Automatic Insight Cards (Dashboard)

The Dashboard generates four AI-powered insight cards automatically (no user input required):

1. **Current Status:** Event count in last 24h + risk level (high / medium / low)
2. **Strongest Event:** Location and magnitude of the most powerful recent earthquake
3. **Top Affected City:** City with highest earthquake activity and max magnitude
4. **Anomaly Alert:** Detects unusual seismic activity spikes compared to 11-hour baseline average

Three AI-generated **action recommendations** are also displayed (e.g., "Activity above normal — stay alert and review your emergency plan").

2.3.3 3. Planned Future Interactions

Planned Features for Future Assignments

- **Address-Based Risk Report Form:** User enters their address → AI returns nearest fault distance, soil class, historical activity summary, and overall risk score
- **Building Safety Assessment Form:** User enters building age, floor count, and construction type → AI provides structural risk evaluation
- **Earthquake Bag Checklist Generator:** Personalized list based on household size, children/elderly/pets
- **Push Notifications:** Alerts when significant earthquakes occur near user's saved location

3 System Architecture (High-Level)

3.1 Architecture Overview

The system follows a three-layer architecture: **Frontend** (Angular 21), **Backend** (Spring Boot 4.0), and **External Services** (Kandilli Observatory, MTA GeoJSON data, Groq AI API). The backend acts as a gateway that aggregates data from multiple sources and provides a unified REST API to the frontend.

3.2 Architecture Diagram

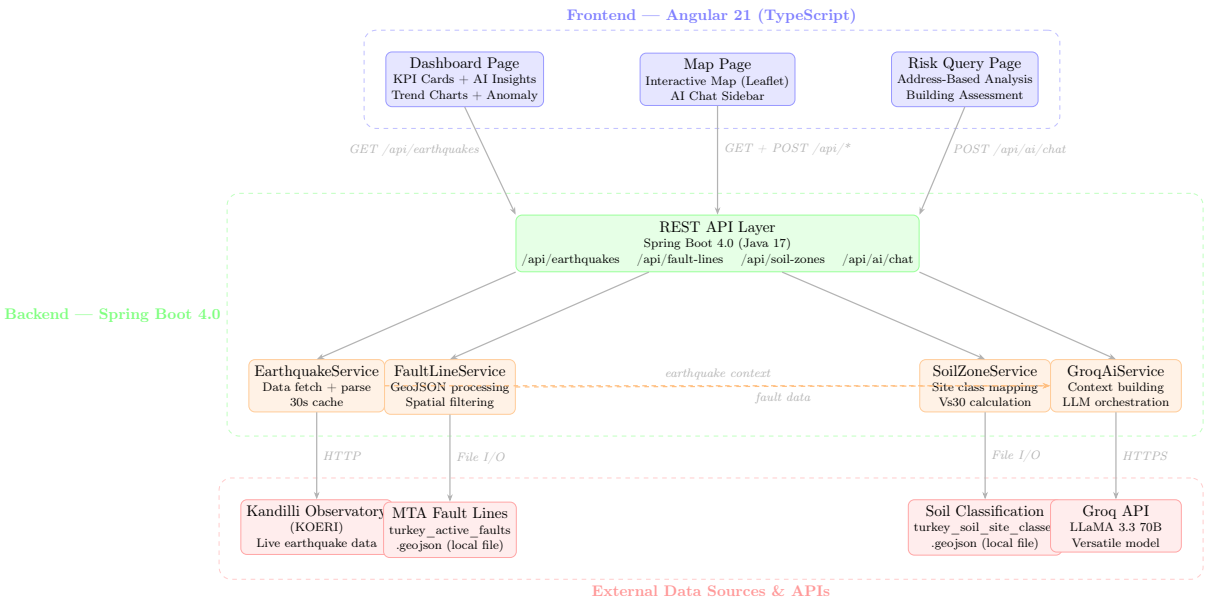

Figure 1: System architecture diagram. The frontend communicates with the backend via REST API. The backend aggregates data from Kandilli Observatory (live earthquakes), local GeoJSON files (fault lines and soil zones), and Groq API (AI analysis). The GroqAiService receives earthquake and fault data as context before calling the LLM.

3.3 AI Agent – Frontend Interaction

The AI Agent interacts with users at two frontend touchpoints:

1. **Map Page — AI Chat Sidebar:**
   - Users type questions in a chat panel on the right side of the map
   - The frontend automatically attaches context to each request: currently selected earthquake (ID, coordinates, magnitude), map bounding box (viewport), and active filters (time window, min magnitude)
   - AI responses are displayed in a chat-style interface with a metadata footer showing model name, event count, and nearest fault distance
   - Quick preset buttons allow one-click queries for common questions
2. **Dashboard — AI Insight Cards:**
   - Runs as **background automation** — no user input required

- Fetches earthquake data, computes statistics, and generates four severity-coded insight cards
- Calculates anomaly score by comparing latest hour's activity against 11-hour baseline
- Produces three actionable recommendations based on current seismic conditions

3.4 AI Agent – Backend Interaction

When a chat request arrives at `POST /api/ai/chat`, the GroqAiService executes the following pipeline:

1. **Data Collection:** Fetches recent earthquakes from EarthquakeService. Applies user filters (hours: 1–168, minMagnitude: 0–10, limit: 1–500, bounding box).
2. **Focus Event Selection:** Identifies the target earthquake by event ID, coordinates, or selects the strongest/most recent event as fallback.
3. **Fault Proximity Analysis:** Calculates distance from focus event to all fault line segments within 35km threshold. Extracts fault characteristics: slip type, last movement date, net slip rate, average dip, and rake angle.
4. **Context Construction:** Builds a structured context string containing: event count, magnitude statistics, focus event details, top 4 nearest fault candidates with distances, fault activity summary (event counts per fault), and 12 most recent events.
5. **LLM Request:** Sends system prompt (Turkish seismic assistant persona) + context + user question to Groq API. Configuration: temperature 0.2, max tokens 700, timeout 25 seconds.
6. **Response Delivery:** Returns AI answer enriched with metadata (model name, event count, nearest fault distance, processing notes).
7. **Fallback:** If the API key is missing or the API call fails, generates a rule-based local summary with distance band classification (very close $\leq$2km, close 2–8km, medium 8–20km, far >20km).

3.5 External APIs and Services

Service	Purpose	Protocol	Update Frequency
Kandilli Observatory (KOERI)	Live earthquake data	HTTP (text parse)	30-second cache
MTA Fault Lines	Active fault geometries	Local GeoJSON file	Static (loaded at startup)
Soil Site Classification	Vs30 / site class zones	Local GeoJSON file	Static (loaded at startup)
Groq API	LLM-based NL analysis	HTTPS REST	On each user request

Table 3: External data sources and services used by the system.

3.6 Technology Stack

Technology	Purpose
Angular 21 (TypeScript 5.9)	Frontend framework with standalone components
Leaflet & MapLibre GL	Interactive map rendering and geospatial layers
RxJS	Reactive data streams and state management
Spring Boot 4.0 (Java 17)	Backend REST API framework
Spring Data JPA + H2	Data persistence layer
Groq API (LLaMA 3.3 70B)	AI natural language analysis
Jackson	JSON serialization and deserialization
Vitest	Frontend unit testing framework

Table 4: Complete technology stack.

4 Future AI Agent Enhancements

The following features are planned for future assignments. The current modular architecture (separate services for earthquakes, faults, soil zones, and AI) allows these to be added incrementally without architectural changes.

Phase 1 — Earthquake Knowledge & Preparedness

- **General Earthquake Information:** AI answers fundamental questions — what causes earthquakes, what magnitude scales mean, depth classification, aftershock patterns, and how seismic waves propagate.
- **Earthquake Bag Checklist:** Personalized emergency bag list based on household size, presence of children/elderly/pets, and regional risk level.
- **During-Earthquake Guide:** Location-specific instructions — what to do at home, in a car, outdoors, at school, in an elevator.
- **Post-Earthquake Actions:** Step-by-step checklist — gas valve check, crack inspection, aftershock awareness, when to evacuate.

Phase 2 — Home & Building Risk Analysis

- **Address-Based Risk Report:** User enters their address → system geocodes it, finds nearest fault (with distance), determines soil classification zone, retrieves historical earthquake data for the area, and generates an overall risk score.
- **Building Safety Assessment:** User provides building age, number of floors, construction type (reinforced concrete, masonry, steel frame) → AI evaluates structural risk considering soil zone and fault proximity.
- **Neighborhood Comparison:** "Is Kadikoy or Bakirkoy safer?" → AI compares fault distances, soil classes, and historical seismicity for both areas.

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5 Summary

Seismic Command Dashboard aims to make earthquake awareness accessible to everyone, not just seismologists. The AI Agent serves as a personal earthquake advisor that can explain what just happened, assess home safety risks, guide preparedness efforts, evaluate building safety, and provide earthquake statistics — all through natural conversation in Turkish.

The current implementation already provides real-time earthquake monitoring, interactive fault line visualization, AI-powered chat with fault-earthquake correlation, and automatic dashboard insights. The three-layer architecture (Angular + Spring Boot + Groq LLM) is designed for incremental feature expansion in upcoming assignments.

Source code: https://github.com/YunusEmre3757/advenced_web.git/